Descriptive Statistics

LLE – Mathematics and Statistics

1. Below is a questionnaire that is given to customers who visit the “Pecs Bar & Gym”, the place for those who want mean pecs, or a drink. Determine the level of data (Nominal, Ordinal, Interval, Ratio) produced for each of the questions.

   a) Do you use a car to come to the gym?

      - Yes
      - No

      Solution

      The answers are distinct categories with no inherent order, so nominal.

   b) How would you rate the cleanliness of our facilities?

      - Excellent
      - Good
      - Average
      - Poor

      Solution

      The answers have a meaningful order, but the intervals between them are not equal, so ordinal.

   c) What distance, in km, do you live from the gym? (Please provide a number)

      Solution

      The data is numerical, has equal intervals, and a true zero point, so ratio.

   d) How many times did you visit the gym last month?

      Solution

      The data is a numerical count with equal intervals and a true zero, so ratio.

   e) Which of these services did you last use?

      - Free Weights
      - Cardio Machines
      - Fitness Class
      - Bar

Solution

The answers are distinct categories with no inherent order, so nominal.

2. A sample of 16 customers are asked about the session they attended on their last visit and whether they came by car. The results are shown below:

Session	Journey
Free Weights	Car
Cardio Machines	Car
Fitness Class	No Car
Bar	No Car
Cardio Machines	Car
Free Weights	No Car
Fitness Class	Car
Cardio Machines	No Car
Bar	No Car
Free Weights	Car
Cardio Machines	Car
Fitness Class	Car
Free Weights	Car
Bar	Car
Cardio Machines	Car
Fitness Class	No Car

a) Create a cross-table (contingency table) for the frequencies of customers, with session in the rows and journey in the columns.

Session/Journey	Car	No Car	Total
Free Weights	**3**	**1**	**4**
Cardio Machine	**4**	**1**	**5**
Fitness Class	**2**	**2**	**4**
Bar	**1**	**2**	**3**
Total	**10**	**6**	16

b) What percentage of the 16 customers used a car?

Solution

From the table, 10 out of 16 customers used a car.

$\frac{10}{16} \times 100 = 62.5$

c) Of those who used a car, what percentage used the cardio machines?

Solution

From the table, 4 customers used cardio machines and a car, out of a total of 10 car users.

$\frac{4}{10} \times 100 = 40$

d) Create a stacked bar chart, with session on the horizontal axis and a vertical axis that represents the percentage of customers for that session. Each bar should be stacked to show the proportion who used a car and who didn't.

Solution

A stacked bar chart, with percentages, puts the proportion of car and no car on top of each other for each of the different gym activities

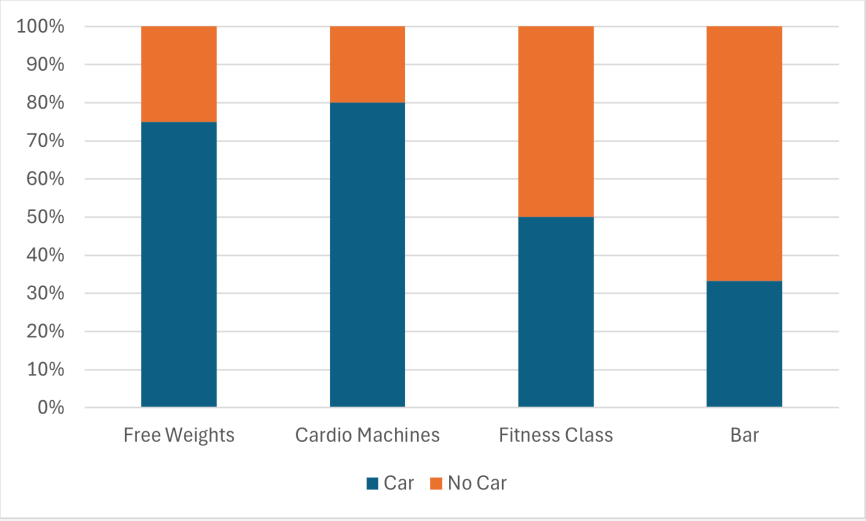
For car percentages:

Free weights 75

Cardio 80

Fitness 50

Bar 33.3



3. The gym is interested in how long it takes people to get to the gym. They conduct a survey of 8 people and ask how long it has taken them to get to the gym in minutes. All 8 people used a car to get to the gym. The sample results were:

12, 15, 16, 18, 20, 21, 22, 28

a) Calculate the mean time in minutes.

Solution

Sum of values $12 + 15 + 16 + 18 + 20 + 21 + 22 + 28 = 152$

Mean = $\frac{152}{8} = 19$

b) Calculate the median time in minutes.

Solution

The data is already ordered. The middle two values are the 4th and 5th: 18 and 20.

The median is the average of these two: $\frac{18+20}{2} = 19$

c) Calculate the range of times in minutes.

Solution

The range is the difference between the minimum and maximum value.

$28 - 12 = 16$

The formula below can be used to calculate the **standard deviation**, s:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Where

$\sum$ is the sum

x_i is each of the data points

$\bar{x}$ is the mean of the data

n is the sample size

d) The table below can be used in calculating the standard deviation. Complete the missing values.

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	-7	49
15	-4	**16**
16	**-3**	**9**
18	**-1**	**1**
20	1	**1**
21	**2**	**4**
22	**3**	**9**
28	**9**	**81**
Sum Σ	—	**170**

e) Calculate the value for the sample standard deviation. Give your answer correct to 2 decimal places.

Solution

$$s = \sqrt{\frac{170}{8-1}} = 4.93$$

f) In a sample of people who did not use a car, the following statistics were found:

- Mean = 25 minutes
- Median = 24 minutes
- Range = 18 minutes
- Standard Deviation = 6.2 minutes

Compare the time taken by car users with the time taken by non-car users to get to the gym.

Solution

Comparison of Averages: Car users are faster on average (Mean 19 vs 25; Median 19 vs 24).
Comparison of Spread: Car users have a more consistent journey time as their range and standard deviation are lower (Range 16 vs 18; SD 4.93 vs 6.2).

4. The following dataset of 10 numbers has a mean of 5.5 and a sample standard deviation of 1.96.

   5, 4, 7, 7, 8, 9, 4, 5, 4, 3

   Showing how you can use the information above, find the mean and the standard deviation of the following datasets.

   a) 5000, 4000, 7000, 7000, 8000, 9000, 4000, 5000, 4000, 3000

   Solution

   Each value has been multiplied by 1000.
   New Mean: $5.5 \times 1000 = 5500$
   New Standard Deviation: $1.96 \times 1000 = 1960$

   b) 105, 104, 107, 107, 108, 109, 104, 105, 104, 103

   Solution

   100 has been added to each value.
   New Mean: $5.5 + 100 = 105.5$
   New Standard Deviation: Remains unchanged by addition and subtraction: 1.96

   c) 5100, 4100, 7100, 7100, 8100, 9100, 4100, 5100, 4100, 3100

   Solution

   Each value has been multiplied by 1000, then 100 has been added.
   New Mean: $5.5 \times 1000 + 100 = 5600$
   New Standard Deviation: Only affected by the multiplication: $1.96 \times 1000 = 1960$

   Use the same technique, showing your working, to find the mean and standard deviation of the following dataset of 8 numbers:

   d) 20000, 20000, 30000, 40000, 60000, 70000, 70000, 90000

Solution

Step 1: Code the data Divide all numbers by 10000:

$$2, 2, 3, 4, 6, 7, 7, 9$$

Step 2: Find the mean of the coded data:

$$\frac{40}{8} = 5$$

Step 3: Find the standard deviation of the coded data:

$$\sqrt{\frac{48}{8-1}} = 2.6186$$

Step 4: Decode the results

Mean = $5 \times 10000 = 50000$

S.D. = $2.6186 \times 10000 = 26186$